

Logistic Regression

Classification and binary outcomes

David Puelz

The University of Austin

The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

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The OkCupid data set

- The OkCupid data set contains information about 59946 profiles from users of the OkCupid online dating service.
- We have data on user age, height, sex, income, sexual orientation, education level, body type, ethnicity, and more.
- OkCupid often publishes their own analyses—see <https://theblog.okcupid.com/tagged/data>.
- Let's see if we can predict the sex/gender of the user based on their height.



What's wrong with this regression?

$$\widehat{\text{sex}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{height}$$



What's wrong with this regression?

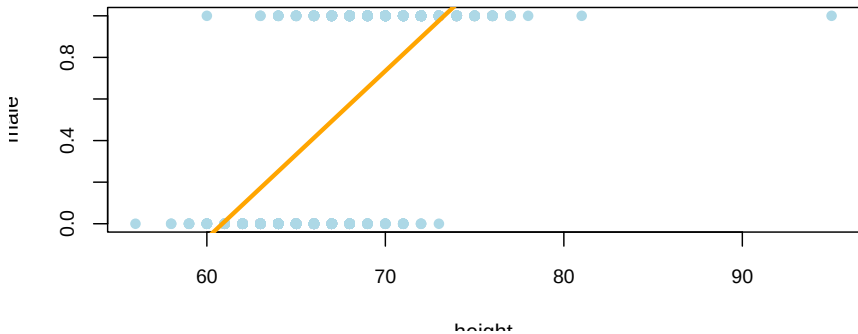
$$\widehat{\text{sex}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{height}$$

The Y variable here is **categorical** (two levels—male or female), so regular linear regression might not be the best choice.



But what if we just do it anyway?

We created a dummy: male = 1, female = 0. Fitting a line:



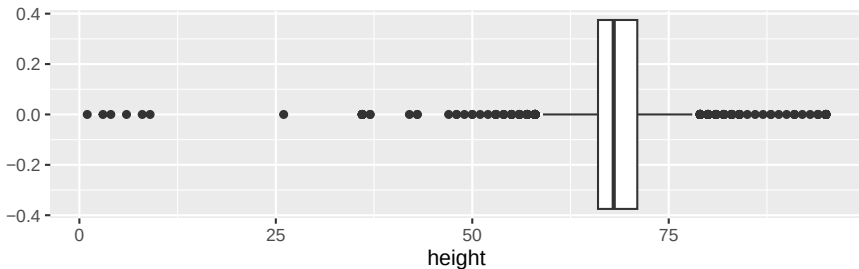
A line is a spectacularly bad fit. And what does it mean to predict male = 0.7 (or 1.2)?



Cleaning the data

There are some weird values for height:

```
ggplot(profiles, aes(x = height)) + geom_boxplot()
```





Cleaning the data

Consider only heights between 55 and 80 inches (4'7" and 6'8"). This excludes only 117 cases out of 59946.

```
my.profiles <- profiles %>% filter(height >= 55 & height <= 80)
```



The idea behind logistic regression

- Instead of predicting male/female, predict the *probability* that they are male.
- One level is “success” ($Y = 1$), the other “failure” ($Y = 0$). Here male = success.
- Fit a curve that is always between 0 and 1.



Odds

- “Even (1/1) odds” $\Rightarrow$ probability of success is $1/2$



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- “Even (1/1) odds” $\Rightarrow$ probability of success is $1/2$
- “2/1 odds” $\Rightarrow$ probability is $2/3$
- In general: odds of something with probability p are $p/(1 - p)$



Odds example

During March Madness, a \$100 bet on San Diego State paid \$360—odds to win of $1/3.6 = 5/18$. Implied *probability* that SDSU wins?



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$$\frac{p}{1-p} = \frac{1}{3.6} \Rightarrow p = \frac{1}{4.6} \approx 0.22$$



The logistic regression model

Logistic regression models the **log odds** of success as a linear function of X :

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

This fits an S-shaped curve (bounded between 0 and 1).



Let's try it

```
model <- glm(male ~ height, data=my.profiles, family=binomial)
summary(model)
```



Let's try it

Call:

```
glm(formula = male ~ height, family = binomial, data = my.profiles)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-44.44861	0.35751	-124	<2e-16 ***
height	0.66190	0.00529	125	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80654 on 59825 degrees of freedom
Residual deviance: 44637 on 59824 degrees of freedom
AIC: 44641

Number of Fisher Scoring iterations: 6



How to interpret the curve?

The regression says

$$\log(\text{odds}) = \log \left(\frac{P(\text{male})}{1 - P(\text{male})} \right) = -44.45 + 0.66 \cdot \text{height}.$$



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The regression says

$$\log(\text{odds}) = \log\left(\frac{P(\text{male})}{1 - P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}.$$

Solving for $P(\text{male})$:

$$\widehat{P(\text{male})} = \frac{e^{-44.45+0.66 \cdot \text{height}}}{1 + e^{-44.45+0.66 \cdot \text{height}}}$$



Making predictions

```
predict(model, data.frame(height=69), type="response")
```

1

0.77

So $\frac{e^{-44.45+0.66 \cdot 69}}{1+e^{-44.45+0.66 \cdot 69}} = 0.77$: we predict someone who is 5'9" has about 77% chance of being male.

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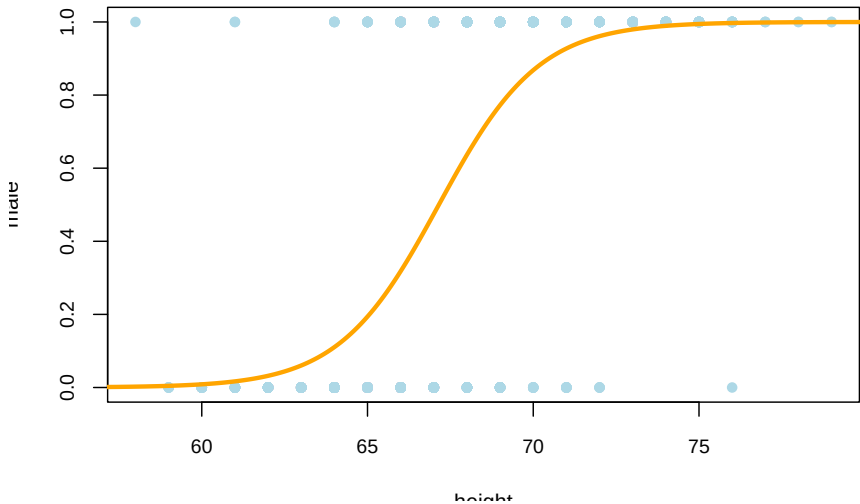
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Visualizing the model





Interpreting the coefficients

$$\log \left(\frac{P(\text{male})}{1 - P(\text{male})} \right) = -44.45 + 0.66 \cdot \text{height}.$$

- When height = 0, predicted log odds = -44.45 ; predicted probability $\approx 0\%$.



Interpreting the coefficients

$$\log \left(\frac{P(\text{male})}{1 - P(\text{male})} \right) = -44.45 + 0.66 \cdot \text{height}.$$

- When height = 0, predicted log odds = -44.45 ; predicted probability $\approx 0\%$.
- When height increases by 1 inch, log odds increase by 0.66.



Interpreting the coefficients

Rewriting: Predicted odds of male = $e^{-44.45+0.66 \cdot \text{height}}$. Increasing height by 1 inch *multiplies* the odds by $e^{0.66} = 1.94$; i.e., increases odds by 94%.



Interpreting the coefficients

Rewriting: Predicted odds of male = $e^{-44.45+0.66 \cdot \text{height}}$. Increasing height by 1 inch *multiplies* the odds by $e^{0.66} = 1.94$; i.e., increases odds by 94%. Increasing height by 2 inches multiplies the odds by $e^{2 \cdot 0.66} = 3.76$; i.e., 276% increase.

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How good is our model?

- The usual R^2 isn't available for logistic regression.



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- The usual R^2 isn't available for logistic regression.
- There are “pseudo- R^2 ” metrics, but they're harder to interpret.



How good is our model?

- The usual R^2 isn't available for logistic regression.
- There are “pseudo- R^2 ” metrics, but they're harder to interpret.
- We'll focus on something simple: how many cases did we get right?



How many cases did we accurately predict?

Rule: predict male if $\widehat{P(\text{male})} \geq 0.5$, else female.

```
predicted.sex <- ifelse(predict(model, type="response") >= 0.5, 1, 0)
correct <- ifelse(predicted.sex == my.profiles$sex, 1, 0)
sum(correct) / nrow(my.profiles)

[1] 0.83
```



Comparing to a baseline

83% accuracy. Compare to always predicting the majority (male):

```
xtabs(~ sex, data=my.profiles) %>% prop.table()
```

```
sex  
  f   m  
0.4 0.6
```

Our model gives a “lift” from 60% (baseline) to 83%.



The confusion matrix

Pick one category as “positive” (here: male) and the other as “negative” (female). Every case falls into: TP, TN, FP, FN.



The confusion matrix

		Actual	
		Positive	Negative
Predicted	Positive	TP	FP
	Negative	FN	TN

TP = true male predicted male; TN = true female predicted female; FP = female predicted male; FN = male predicted female.



The confusion matrix

```
tab_conf <- xtabs(~ predicted.sex + my.profiles$sex)
tab_conf
```

	my.profiles\$sex	
predicted.sex	f	m
f	19466	5494
m	4623	30243

```
fn <- tab_conf[1,2]
```

```
fp <- tab_conf[2,1]
```

5494 false negatives (true M predicted F) and 4623 false positives (true F predicted M).



Rates from the confusion matrix

- True positive rate = $TP/(TP+FN)$: proportion of actual positives predicted positive
- True negative rate = $TN/(TN+FP)$: proportion of actual negatives predicted negative
- False positive rate = $FP/(TN+FP)$; False negative rate = $FN/(TP+FN)$



Confusion matrix (normalized)

```
xtabs(~ predicted.sex + my.profiles$sex) %>% prop.table(2)
```

	my.profiles\$sex	
predicted.sex	f	m
f	0.81	0.15
m	0.19	0.85

Use `predicted ~ actual`; `prop.table(2)` normalizes by column (actual).



Model evaluation: final thoughts

- You can trade off FP vs FN by changing the threshold (predict 1 when $\hat{p} > p$ for $p \neq 0.5$).
- FP and FN can have very different costs (e.g., loans, medical tests).
- These are **in-sample** measures—optimistic. Use train/test or cross-validation for out-of-sample accuracy.

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Recap and adding predictors

Single-predictor model: $\log(\text{odds}) = -44.45 + 0.66 \cdot \text{height}$. We can add predictors just like in linear regression. Each coefficient is the effect on *log odds* (or odds) holding others constant.

```
model2 <- glm(male ~ height + orientation, data=my.profiles, family=binomial)
```



Adding another predictor

orientation has three categories:

```
xtabs(~ orientation, data=my.profiles)
```

```
orientation
```

bisexual	gay	straight
2763	5568	51495

Call:

```
glm(formula = male ~ height + orientation, family = binomial,  
     data = my.profiles)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-46.08076	0.37167	-124.0	<2e-16 ***
height	0.66535	0.00537	124.0	<2e-16 ***
orientationgay	2.09556	0.07209	29.1	<2e-16 ***
orientationstraight	1.39972	0.06068	23.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80654 on 59825 degrees of freedom
Residual deviance: 43722 on 59822 degrees of freedom
AIC: 43730

Number of Fisher Scoring iterations: 6



Interpreting coefficients (multiple predictors)

$$\log \left(\frac{p}{1-p} \right) = -46.08 + 0.67 \cdot \text{height} + 2.1 \cdot \text{gay} + 1.4 \cdot \text{straight}.$$

- Intercept: predicted log odds for bisexual, height 0.



Interpreting coefficients (multiple predictors)

$$\log \left(\frac{p}{1-p} \right) = -46.08 + 0.67 \cdot \text{height} + 2.1 \cdot \text{gay} + 1.4 \cdot \text{straight}.$$

- Intercept: predicted log odds for bisexual, height 0.
- Among same orientation, each extra inch of height multiplies odds by $e^{0.67} = 1.95$ (95% increase).



Interpreting coefficients (dummies)

- Same height, gay vs bisexual: odds multiplied by $e^{2.1} = 8.13$ (713%).



Interpreting coefficients (dummies)

- Same height, gay vs bisexual: odds multiplied by $e^{2.1} = 8.13$ (713%).
- Same height, straight vs bisexual: odds multiplied by $e^{1.4} = 4.05$ (305%).



Understanding what's going on

```
xtabs(~ sex + orientation, data=my.profiles)
```

	orientation		
sex	bisexual	gay	straight
f	1994	1586	20509
m	769	3982	30986

```
xtabs(~ sex + orientation, data=my.profiles) %>% prop.table(2)
```

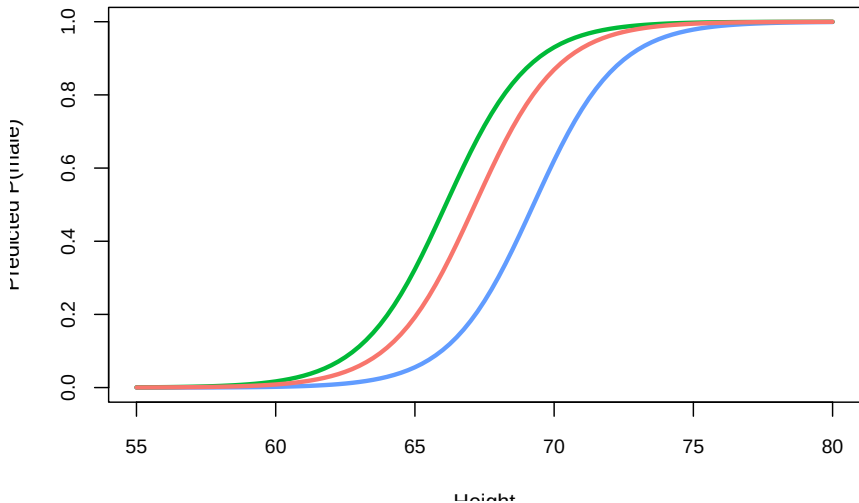
	orientation		
sex	bisexual	gay	straight
f	0.72	0.28	0.40
m	0.28	0.72	0.60





Three curves by orientation

Straight (red), gay (green), bisexual (blue):



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Interactions

In linear regression, an interaction means the **slope** of X_1 depends on the **value** of X_2 . Same in logistic regression: the effect of X_1 on the *odds* depends on X_2 . Try height $\times$ orientation.

```
int.model <- glm(male ~ height * orientation, data=my.profiles, family=binomial)
summary(int.model)
```

Call:

```
glm(formula = male ~ height * orientation, family = binomial,
     data = my.profiles)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-35.3027	1.4050	-25.13	< 2e-16	***
height	0.5076	0.0206	24.67	< 2e-16	***
orientationgay	-6.2727	1.8365	-3.42	0.00064	***
orientationstraight	-10.2887	1.4596	-7.05	1.8e-12	***
height:orientationgay	0.1218	0.0271	4.49	7.1e-06	***
height:orientationstraight	0.1712	0.0214	8.01	1.2e-15	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80654 on 59825 degrees of freedom
Residual deviance: 43663 on 59820 degrees of freedom
AIC: 43675

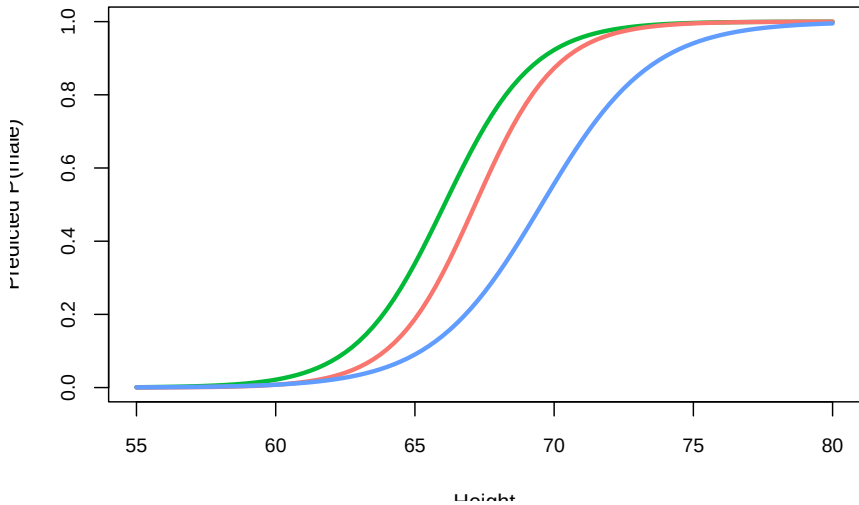
Number of Fisher Scoring iterations: 6



The interaction model

$$\log(p/(1-p)) = -35.3 + 0.51 \cdot \text{height} + -6.27 \cdot \text{gay} + -10.29 \cdot \text{straight} + 0.12 \cdot \text{height} \cdot \text{gay} + 0.17 \cdot \text{height} \cdot \text{straight}.$$

Gay (green), straight (red), bisexual (blue)—different slopes:



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Hypotheses to test

1. **Overall:** $\beta_1 = \beta_2 = \dots = 0$ (no predictive power)



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1. Overall: $\beta_1 = \beta_2 = \dots = 0$ (no predictive power)
2. One quantitative predictor: $\beta_i = 0$ (no effect of that X , holding others constant)



Hypotheses to test

1. **Overall:** $\beta_1 = \beta_2 = \dots = 0$ (no predictive power)
2. **One quantitative predictor:** $\beta_i = 0$ (no effect of that X , holding others constant)
3. **One dummy:** $\beta_i = 0$ (no difference vs reference level, holding others constant)



Overall null: likelihood ratio test

The likelihood ratio test compares the full model to an intercept-only model. Here: $\chi^2 = 36932$, $p < 0.001$ (highly significant).



Testing one coefficient

For a quantitative variable (e.g., height) or a dummy (e.g., straight vs bisexual), use the z and p -value for that coefficient in `summary(model2)`.

```
summary(model2)
```

```
orientationstraight    1.39972    0.06068    23.1    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 80654  on 59825  degrees of freedom
Residual deviance: 43722  on 59822  degrees of freedom
```

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What else can we use logistic regression for?

- **Finance:** Predicting which customers are likely to default on a loan
- **Advertising:** Predicting response to a campaign
- **Marketing:** Predicting purchase or sign-up